



COMMUNICATION IN ENGLISH II



INDUCTIVE AND DEDUCTIVE REASONING

CHINYERE OMENI-NZEWUIHE

M.A English Studies



LEARNING OBJECTIVES

At the end of the lesson, you should be able to

- Elucidate the difference between deductive and inductive reasoning, including recognising their defining characteristics.
- Identify real-world examples of deductive and inductive reasoning and analyse which type of reasoning is being used.
- Apply deductive and inductive reasoning in various contexts.

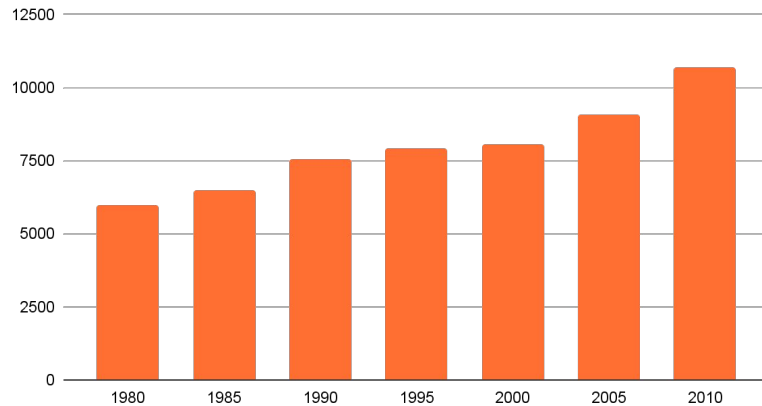




INDUCTIVE VS. DEDUCTIVE REASONING

A village wishes to estimate the population of for the next 15 years. Which reasoning will be appropriate to use, inductive or deductive?

Female Population Growth





WHAT DOES INDUCTIVE REASONING MEAN?

Inductive reasoning is a form of reasoning in which the conclusion is considered likely or probable, but not certain, based on the premises.

Simply put, in inductive reasoning, the method here is to look for patterns so that you can draw conclusions or make a generalisation.





WHAT DOES DEDUCTIVE REASONING MEAN?

Deductive reasoning is a form of reasoning in which the conclusion is necessarily true if the premises are true. In other words, if the premises are true, the conclusion must also be true.

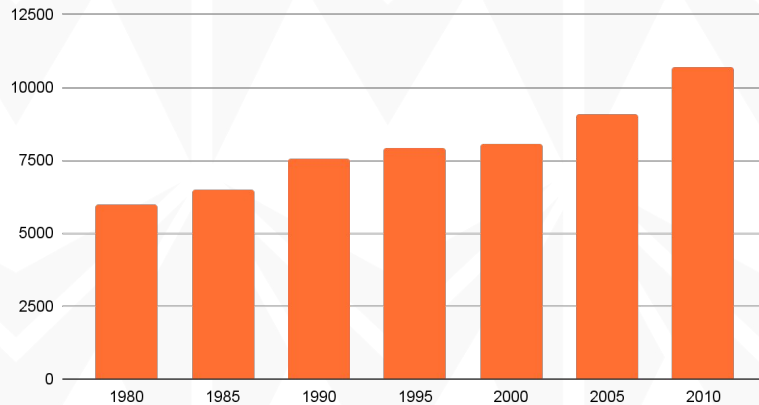
Simply put, deductive reasoning is using a set of data to make a specific conclusion that comes up to be a fact.





WHICH REASONING WILL BE USED?

Female Population Growth



A village wishes to estimate the population of for the next 15 years. Which reasoning will be appropriate to use, inductive or deductive?



INDUCTIVE REASONING

Inductive reasoning is a form of reasoning in which the conclusion is considered likely or probable, but not certain, based on the premises.

Inductive arguments are used to generalise from specific observations or past experiences.





INDUCTIVE REASONING

Inductive reasoning are based on patterns or generalisations. They involve moving from specific instances to a probable or likely conclusion.

For example

Stage	Example
Specific observation	My swan is white.
Pattern recognition (premise)	Every observed swan is white.
General conclusion	Therefore, all swans are white



INDUCTIVE REASONING

Probability: Inductive reasoning provide probable, but not certain, conclusions. The strength of an inductive argument depends on the quantity and quality of the evidence.

Example:

Based on historical weather data, you might conclude that it's likely to be cold in the winter, but you can't be certain of the weather on a specific day.





PROBABILITY OF CONCLUSIONS - INDUCTIVE REASONING

Inductive reasoning provides conclusions that are probable or likely, but not guaranteed to be true. The strength of the conclusion depends on the quantity and quality of the evidence.

Explanation: In this example, inductive reasoning is used to predict a future event (disease outbreak) based on observed patterns (correlation between mosquito activity and disease cases).

Case: Prediction of Disease Outbreak	
Premise 1	In the past, every time there was an increase in mosquito activity, there was a corresponding increase in cases of a mosquito-borne disease.
Premise 2	There has been a recent increase in mosquito activity.
General conclusion	It is likely that there will be a disease outbreak.



TYPES OF INDUCTIVE REASONING

There are various types of inductive reasoning, but we will observe some common types.

Type: Generalisation is a common form of inductive reasoning where you make a broad statement about a group or category based on specific observations.

Example: After observing multiple instances of people wearing sunglasses and hats on sunny days, you might generalise that "*Many people wear sunglasses and hats on sunny days.*"





TYPES OF INDUCTIVE REASONING

Analogical Reasoning

Type: Analogical reasoning is when you draw conclusions about one situation or object based on its similarities to another situation or object.

Example: Since cats are known to be good hunters, and this new breed of cat has the same characteristics, it's likely to be a good hunter as well.



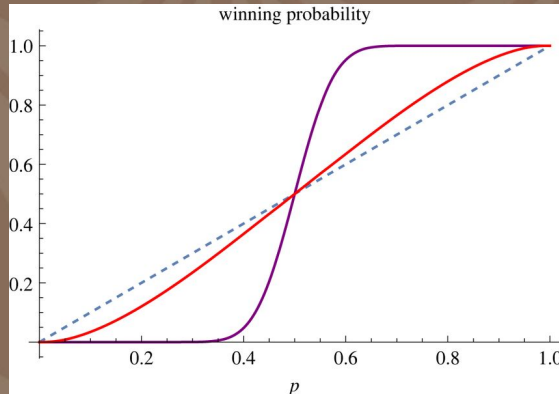


TYPES OF INDUCTIVE REASONING

Statistical Reasoning

Type: Statistical reasoning involves drawing conclusions based on statistical data or probability.

Example: Based on past election results and current polling data, Candidate A has a 70% probability of winning the upcoming election.





TYPES OF INDUCTIVE REASONING

Find out more about these type of inductive reasoning.

- Causal inference
- Prediction
- Abductive reasoning
- Enumerative induction





DEDUCTIVE ARGUMENTS

Deductive reasoning is a form of reasoning in which the conclusion is necessarily true if the premises are true. In other words, if the premises are true, the conclusion must also be true. Deductive arguments are often used in mathematics and formal logic.

Structure: Deductive arguments have a specific structure, where the premises lead to a conclusion based on the rules of logic. The most famous example is a syllogism, like:

All humans are mortal. (Premise)

Socrates is a human. (Premise)

Therefore, Socrates is mortal. (Conclusion)



DEDUCTIVE ARGUMENTS

Certainty of Conclusions

Deductive reasoning provides conclusions that are certain or guaranteed to be true when the premises are true, assuming that the argument is valid.





DEDUCTIVE ARGUMENTS

Example: Categorical Syllogism

Scenario: A classic example of a categorical syllogism.

Premise 1	Premise 2	Conclusion
All mammals have a backbone.	Whales are mammals.	Therefore, whales have a backbone.

Explanation: This is a deductive argument where the premises establish a necessary relationship between mammals and having a backbone, and between whales and being mammals. Therefore, the conclusion is certain, given that the premises are true.





DEDUCTIVE ARGUMENTS

Example: Modus Ponens

Scenario: Another deductive argument form known as Modus Ponens.

Premise 1	Premise 2	Conclusion
If it's raining, then the ground is wet.	It is raining.	Therefore, the ground is wet.

Explanation: Modus Ponens demonstrates that if you affirm the antecedent (the "if" part of a conditional statement) and the conditional statement is true, then the consequent (the "then" part) must be true. This is a valid and certain deductive argument.





DEDUCTIVE ARGUMENTS

Example: Mathematical Proof

Scenario: Demonstrating a mathematical proof.

Premise 1	Premise 2	Conclusion
The sum of two even numbers is always even.	2 and 4 are even numbers.	Therefore, the sum of 2 and 4 is even.

Explanation: In mathematics, deductive reasoning is commonly used to prove theorems and establish mathematical truths based on axioms and established rules. The conclusion here is certain because it follows the logical rules of mathematics.



TYPES OF DEDUCTIVE REASONING

Modus Tollens

Description: Modus Tollens is another valid deductive argument that denies the consequent. It follows the pattern "If A, then B; Not B; therefore, Not A."

Example: "If it's raining, the ground is wet. The ground is not wet. Therefore, it's not raining."





DEDUCTIVE VS INDUCTIVE

Deductive Reasoning

- How can we determine if an argument is deductive, and what criteria must it meet to be considered valid and certain?

Inductive Reasoning

- What are some potential limitations or weaknesses in relying on inductive reasoning when drawing conclusions based on observations and patterns?





DEDUCTIVE VS INDUCTIVE

Deductive Reasoning

- How can we determine if an argument is deductive, and what criteria must it meet to be considered valid and certain?

To determine if an argument is deductive, we should look for a clear structure where the conclusion necessarily follows from the premises. A deductive argument is considered valid if the conclusion follows logically from the premises, and it is sound if the premises are true. Valid deductive arguments guarantee certainty in the conclusions.





DEDUCTIVE VS INDUCTIVE

Inductive Reasoning

- What are some potential limitations or weaknesses in relying on inductive reasoning when drawing conclusions based on observations and patterns?

Inductive reasoning is based on probabilities and generalizations from observed patterns, so it does not guarantee the truth of conclusions. Some limitations and weaknesses of inductive reasoning include the possibility of sampling bias, the reliance on incomplete data, and the potential for coincidental patterns. Additionally, inductive conclusions are contingent on the quality and quantity of the evidence, which can vary.



SUMMARY

In summary, deductive reasoning provides certainty in conclusions when the argument is valid and the premises are true, while inductive reasoning offers probable conclusions based on patterns, evidence, and generalizations. Both forms of reasoning have their strengths and are applied in different contexts based on the nature of the problem or question at hand. Deductive reasoning is akin to "all or nothing" logic, while inductive reasoning acknowledges the nuances and probabilities inherent in many real-world situations.





FURTHER READING & REFERENCES

- Smith, J. A. (2019). The Art of Deductive Reasoning. Logic Press.
- Johnson, M., & Brown, L. K. (2020). Inductive Logic: A Comprehensive Guide. Academic Publishing.





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